FIT2014 Practice Exam (seed 42)

FIT2014 Theory of Computation · Total marks: 120 · Time: 3 hours

Instructions: answer all questions. Where an answer format is suggested, using it lets your answer be marked automatically (`examx grade`).

Question 1 (4 marks)

Your phone is somewhere in your house. Suppose we have propositions B, G, C, L with the following meanings.

- B: Your phone is in the bedroom.
- G: Your phone is in the garage.
- C: Your phone is in the bathroom.
- L: Your phone is in the laundry.

Use B, G, C, L to write a proposition, in Conjunctive Normal Form, that is True precisely when at least one of these 4 propositions is True.

> Answer format for automatic marking: use ∧ ∨ ¬ (or & | ~) and parentheses.

Question 2 (4 marks)

Consider the regular expression (ba∪a*)(ba∪ε) over the alphabet {a,b}.

(a) Write down all strings of at most 6 letters that match this regular expression. *[2 marks]*

(b) Give an NFA that recognises the language described by this regular expression. *[2 marks]*

> Answer format for automatic marking: an edge list — `start: 1`, `final: 3`, then one transition per line such as `1 a 2` or `2 ε 3` (several letters may be comma-separated: `1 a,b 2`).

Question 3 (4 marks)

Consider the regular expression ((ab)∪*(bab))* over the alphabet {a,b}.

(a) Write down all strings of at most 6 letters that match this regular expression. *[2 marks]*

(b) Give an NFA that recognises the language described by this regular expression. *[2 marks]*

> Answer format for automatic marking: an edge list — `start: 1`, `final: 3`, then one transition per line such as `1 a 2` or `2 ε 3` (several letters may be comma-separated: `1 a,b 2`).

Question 4 (3 marks)

A language over alphabet {a,b} is said to be cofinite if it contains all strings over that alphabet except for some finite number of strings. Prove that every cofinite language is regular.

Question 5 (7 marks)

The following table describes an NFA with ε-transitions (each cell lists the set of possible next states). Convert it to a Finite Automaton that recognises the same language, using the subset

construction. Represent your FA as a transition table.

```
       state  a      b    ε
Start  1      {3,5}  ∅    ∅
       2      ∅      {3}  ∅
       3      ∅      {4}  {2}
Final  4      {4}    ∅    ∅
       5      {5}    {5}  ∅
```

Answer format for automatic marking: a transition table whose first row is `state a b`, one row per state, rows prefixed by `Start` and/or `Final`, cells holding a state name (use ∅ for none). Alternatively an edge list: `start: 1`, `final: 2 3`, then lines like `1 a 2`.

Question 6 (3 marks)

Give a regular expression for the language of all strings over {a,b} that contain aba as a substring.

Answer format for automatic marking: use `∪` (or `|`) for union, `*` for Kleene star, `ε` for the empty string.

Question 7 (6 marks)

Consider the following Regular grammar: S → ε | aT | bU (1) T → aT | bS (2) U → aS (3)

(a) Give (i) a derivation, and (ii) a parse tree, for the string baab . *[2 marks]*

(b) Find the Finite Automaton for the language defined by the above grammar. *[2 marks]*

(c) Give a regular expression for the language defined by the above grammar. *[2 marks]*

Question 8 (2 marks)

Give a Context-Free Grammar for the language { a^m b^n : m < n }.

Answer format for automatic marking: one production per line, e.g. `S → aSb | ε`; non-terminals are capital letters and S is the start symbol.

Question 9 (6 marks)

Consider the following Context-Free Grammar S → a S (1) S → S b (2) S → ε (3) Prove by induction that every string of the form $a^m b^n$, where m ≥ 0 and n ≥ 0, can be generated by this grammar.

Question 10 (5 marks)

Consider the following Context-Free Grammar. S → aBa (1) B → BB (2) B → Q (3) B → R (4) Q → q (5) R → r (6) Give

(a) a derivation, and *[3 marks]*

(b) a parse tree, for the string aqrqa, labelling each step in the derivation on its right by the number of the rule used. Use the spaces below for your answers. (a) Derivation: (b) Parse tree: *[2 marks]*

Question 11 (5 marks)

A string over the alphabet {0,+,–} is said to be balanced if it satisfies both the following: • (i) for any i, the first i characters in the string contain at least as many + as –; • (ii) the whole string has the same number of + as –. Give a Context-Free Grammar for the language.

Question 12 (4 marks)

State two important results that can be proved using the Chomsky Normal Form for Context- Free Grammars.

Question 13 (6 marks)

Draw a Turing machine that adds 1 to a binary number (most significant bit first). If the input is all 1s, the output is one bit longer, e.g. 111 becomes 1000. The input is a string over {0,1}, starting in the leftmost cell with the head on it. For example, on input 1011 the output is 1100.

> Answer format for automatic marking: `start: 1`, `halt: H`, then one transition per line written `state read -> next write move` (move is L, R or S; Δ is the blank), e.g. `1 a -> 2 b R`.

Question 14 (4 marks)

(a) State the Church-Turing thesis. *[2 marks]*

(b) Give two reasons why the Church-Turing thesis is widely accepted. *[2 marks]*

Question 15 (4 marks)

For each of the following decision problems, indicate whether it is Decidable or Undecidable.

(a) Input: a Turing machine M.
Question: Does M ever write the symbol a on its tape, when run on the empty string, within its first 1000 steps?

(b) Input: a Turing machine M and a string w.
Question: If M is run on w, does it halt within 100 steps?

(c) Input: a Turing machine M.
Question: Is the language accepted by M finite?

(d) Input: a Turing machine M.
Question: Does M halt when given the empty string as input?

(a)

(b)

(c)

(d)

Question 16 (10 marks)

Consider the classes of languages Finite $\subset$ Regular $\subset$ Context-Free $\subset$ P $\subseteq$ NP $\subset$ Decidable $\subset$ Recursively Enumerable (r.e.). For each language below, state the **smallest** of these classes that it is known to belong to, or **None** if it is not r.e. (Assume P $\neq$ NP.)

(a) The set of all graphs containing no circuit.
(b) The set of all correctly formed arithmetic expressions using positive integers, +, – and parentheses.
(c) The set of all (encodings of) Turing machines that accept every string.
(d) The set of binary representations of prime numbers.
(e) The language of all strings over {a,b} that do not contain bb as a substring
(f) The language of positive integers in binary with leading bit 1
(g) The language of binary strings (possibly with leading zeros) that represent a multiple of 3
(h) The set of all connected graphs.
(i) The language { ww : w ∈ {a,b}^* }
(j) The language of strings over {a,b} with more a's than b's

(a)

(b)

(c)

(d)

(e)

(f)

(g)

(h)

(i)

(j)

Question 17 (7 marks)

Use the Pumping Lemma for Context-Free Languages to prove that the language $\{ a^n b^n c^n : n \geq 0 \}$ is not context-free.

Question 18 (5 marks)

For each of the following decision problems, indicate whether it is Decidable or Undecidable.

(a) Input: two Turing machines M1 and M2.
Question: Do M1 and M2 accept the same language?

(b) Input: a Turing machine M.
Question: Does M ever move its head left, when run on the empty string, during its first 50 steps?

(c) Input: a Context-Free Grammar G and a string w.
Question: Does G generate w?

(d) Input: two Finite Automata A and B.
Question: Do A and B accept the same language?

(e) Input: a Turing machine M.
Question: Is the language accepted by M finite?

(a)

(b)

(c)

(d)

(e)

Question 19 (4 marks)

Consider the classes of languages Finite $\subset$ Regular $\subset$ Context-Free $\subset$ P $\subseteq$ NP $\subset$ Decidable $\subset$ Recursively Enumerable (r.e.). For each language below, state the **smallest** of these classes that it is known to belong to, or **None** if it is not r.e. (Assume P $\neq$ NP.)

(a) The set of all Context-Free Grammars that generate the empty language.
(b) The language of binary strings (possibly with leading zeros) that represent a multiple of 3
(c) The language of all strings over {a,b} that do not contain bb as a substring
(d) The language of all strings over {a,b} whose first and last letters are different

(a)

(b)

(c)

(d)

Question 20 (4 marks)

For each of the following decision problems, indicate whether or not it is decidable. You may assume that, when Turing machines are encoded as strings, this is done using the Code-Word Language (CWL). Decision Problem your answer (tick one box in each row) Input: Turing machines M and N. Question: Are the encoded forms of M and N identical? Decidable Undecidable Input: Turing machines M and N. Question: Do M and N have the same time complexity? Decidable Undecidable Input: a Turing machine M. Question: Does M correctly determine whether or not its input string is a palindrome? Decidable Undecidable Input: a Turing machine M, and a string w. Question: Does M ever change any letter of w on the tape? Decidable Undecidable

Question 21 (2 marks)

Prove that the class of regular languages is a subset of the class P.

Question 22 (21 marks)

Consider the language NO MONOCHROMATIC TRIANGLE, which consists of all graphs G such that we can assign colours to the edges of the graph so that (i) each edge is either Black or White, and (ii) there is no triangle (i.e., cycle of length 3) in the graph which is monochromatic (i.e., all its edges have the same colour).

(a) Prove that the language NO MONOCHROMATIC TRIANGLE is in NP. Now, let W be the following graph. b y y

c a e

y y d *[4.0 marks]*

(b) Construct a Boolean expression E_{W} in Conjunctive Normal Form such that the satis- fying truth assignments for E_{W} correspond to solutions to the NO MONOCHROMATIC TRIANGLE problem on the above graph W (i.e., they correspond to colourings of the edges of W which have no monochromatic triangle). *[6.0 marks]*

(c) Give a polynomial-time reduction from NO MONOCHROMATIC TRIANGLE to SAT-ISFIABILITY. *[7.0 marks]*

(d) State the Cook-Levin Theorem. *[2.0 marks]*

(e) Given the facts stated so far in this question: What else, if anything, would you need to prove, in order to show that NO MONOCHRO- MATIC TRIANGLE is NP-complete? *[2.0 marks]*